

Problem 4

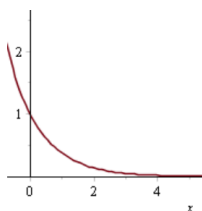
Problem 4

Problem 1

(a) You are given that $f''(x) > 0$ for all x . Which of the following must be true about $f(x)$ on the region $0 \leq x \leq 2$?

- (i) There is a critical point between 0 and 2.
- (ii) There is a local maximum, but not enough information is given to
- (iii) f

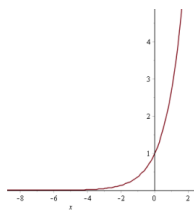
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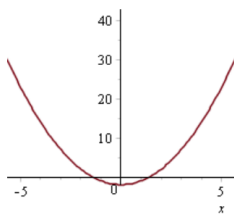
(b) You are told that $f''(x) > 0$ for all x . Which of the following must be true about the graph of $y = f(x)$?

- (i) The graph is a straight line.
- (ii) The graph crosses the x -axis at most once.
- (iii) The graph is concave down.
- (iv) The graph crosses the y -axis more than once.
- (v) The graph is concave up.

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Part (iii) is clearly false since $f''(x) > 0$ means that f is concave up. Part (iv) is false for any function.